Cardiovascular Revascularization Medicine 33 (2021) 1–6

Contents lists available at ScienceDirect

Cardiovascular Revascularization Medicine

Clinical

Transcatheter Aortic Valve Replacement in Low-Risk Bicuspid and Tricuspid Patients: Meta-Analysis

Giorgio A. Medranda [a], Toby Rogers [a,b], Gheorghe Doros [a], Cheng Zhang [a], Brian C. Case [a], Charan Yerasi [a], Christian C. Shults [c], Ron Waksman [a,*]

[a] Section of Interventional Cardiology, MedStar Washington Hospital Center, Washington, DC, United States of America
[b] Cardiovascular Branch, Division of Intramural Research, National Heart, Lung and Blood Institute, National Institutes of Health, Bethesda, MD, United States of America
[c] Department of Cardiac Surgery, MedStar Washington Hospital Center, Washington, DC, United States of America

ARTICLE INFO

Article history:
Received 23 June 2021
Accepted 23 June 2021

Keywords:
Bicuspid aortic stenosis
Low risk
Transcatheter aortic valve replacement

ABSTRACT

Background: Most pivotal transcatheter aortic valve replacement (TAVR) trials have excluded patients with bicuspid aortic stenosis (AS). This study compared TAVR in low-risk patients with bicuspid AS to those with tricuspid AS, incorporating data from prospective trials.
Methods: We selected prospective US low-risk TAVR trials containing a bicuspid arm for this meta-analysis, examining outcomes at 30 days. Study results were pooled using a hierarchical Bayesian random-effects model.
Results: Included were 3 Food and Drug Administration (FDA)-approved investigational device exemption (IDE) trials that enrolled a total of 1810 low-risk patients with symptomatic severe AS for TAVR. We compared 380 bicuspid patients with 1430 tricuspid patients. Event rates at 30 days overall were low, with similar mortality (odds ratio [OR], 0.38; 95% credible interval [CrI]: 0.08, 1.78; I^2, 0%), non-disabling stroke (OR, 0.45; 95% CrI: 0.15, 1.07; I^2, 9%), and permanent pacemaker implantation (OR, 0.86; 95% CrI: 0.41, 1.47; I^2, 59%). There were statistically significant differences in disabling stroke (OR, 0.16; 95% CrI: 0.01, 0.90; I^2, NA) and coronary obstruction (OR, 0.21; 95% CrI: 0.05, 0.91) that disappeared after sensitivity analysis by adding a single event to the tricuspid arm. Hemodynamics were similar at 30 days.
Conclusions: Preliminary data from the FDA-approved IDE trials of low-risk patients with bicuspid AS undergoing TAVR demonstrated 30-day outcomes comparable to low-risk tricuspid patients, except for a trend toward higher stroke in bicuspid patients. Randomized trials are warranted to reassure the safety and long-term outcome of TAVR in patients with severe bicuspid AS.

© 2021 Published by Elsevier Inc.

1. Introduction

Randomized clinical trials comparing transcatheter aortic valve replacement (TAVR) to surgical aortic valve replacement (SAVR) in low-risk patients with severe tricuspid aortic stenosis (AS) have demonstrated comparable outcomes up to 2 years [1,2]. Data on the performance of TAVR in low-risk patients with severe bicuspid AS are sparse, as these patients were excluded from the randomized clinical trials. We sought to compare the safety and efficacy of TAVR in low-risk patients with bicuspid AS to those with tricuspid AS, incorporating the totality of available data from prospective trials.

Abbreviations: AS, aortic stenosis; AVA, aortic valve area; CrI, credible interval; FDA, Food and Drug Administration; MG, mean gradient; OR, odds ratio; PVL, paravalvular leak; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement.

* Corresponding author at: MedStar Washington Hospital Center, 110 Irving St., NW, Suite 4B-1, Washington, DC 20010, United States of America.
E-mail address: ron.waksman@medstar.net (R. Waksman).

2. Materials and methods

We identified 3 prospective trials that met our inclusion criteria (Fig. 1, Table 1). These trials compared TAVR to SAVR for the treatment of symptomatic severe tricuspid AS in low-risk patients [1–3]. Additionally, these trials all included enrollment of bicuspid patients undergoing TAVR into a separate arm [4–6]. Published trials fulfilling the criteria above were identified from an electronic search of MEDLINE (last searched on December 31, 2020). Key search words included: "transcatheter aortic valve replacement" and "low-risk." The search and data collection were performed by 2 authors (G.A.M., B.C.C.) according to the preferred reporting items for systematic review and meta-analysis (PRISMA) guidelines [7]. Prospective trials that met criteria and contained a prospective bicuspid TAVR registry were selected for systematic review and meta-analysis. We preregistered our study with PROSPERO (ID: 233757).

We examined disabling stroke, all-cause mortality, non-disabling stroke, major vascular complications, permanent pacemaker implantation, and coronary obstruction at 30 days in bicuspid patients

https://doi.org/10.1016/j.carrev.2021.06.123
1553-8389/© 2021 Published by Elsevier Inc.

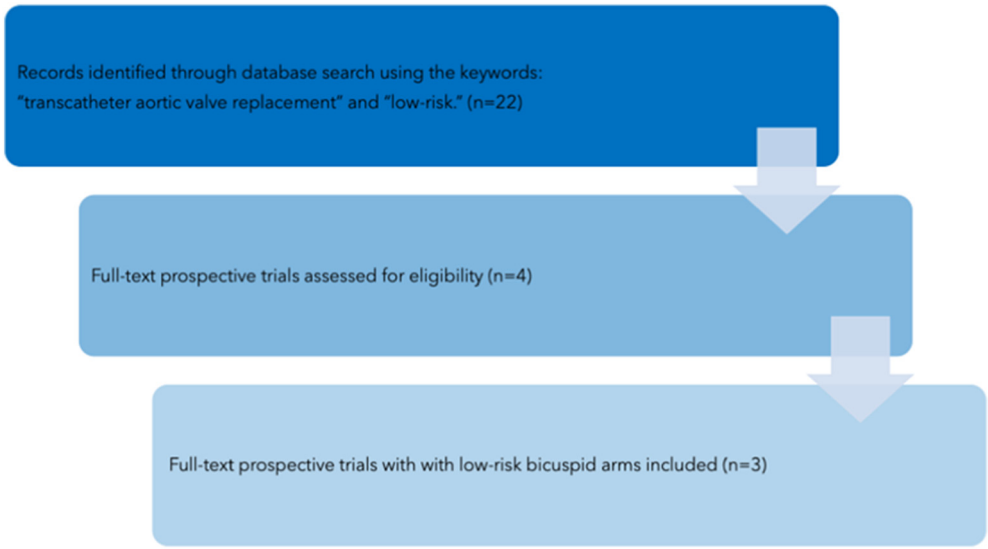

Fig. 1. Study selection.

undergoing TAVR, comparing them to tricuspid patients undergoing TAVR. Additionally, we examined haemodynamics at 30 days, including aortic valve mean gradient (MG), aortic valve area (AVA), and > mild paravalvular leak (PVL). All clinical endpoints were adjudicated by independent clinical events adjudication committees within each trial [1–3].

We included studies that were heterogeneous with respect to study intervention and trial methods. To accommodate the heterogeneity across studies due to valve used, we used a Bayesian random-effects meta-analysis to synthesize results. In particular, we used a Bayesian hierarchical model to estimate the random-effects model. The open-source statistical software for Bayesian analysis with JAGS (Just Another Gibbs Sampler) version 4.3.0 was used to fit the models. The analyses were conducted from R using the 'rjags' [8]. Posterior summaries (point estimated for the meta-analytic odds ratio [OR], mean difference [MD], and 95% credible intervals [CrIs] – the Bayesian equivalent of standard confidence intervals) were calculated by sampling from the posterior distribution of the parameters using Markov Chain Monte Carlo approaches. For each analysis, 2 million samples with a thin of 20 after excluding 10 thousands samples were used. We used non-informative priors (normal distribution with mean = 0, standard deviation = 100) for the hyper parameter of mean (for OR and MD) and half-Chauchy for the between-study standard deviation. To account for the potential large between-study variability and provide for a converging statistical model, a bound of 2 was imposed on the between-study standard deviation with an *a priori* probability of 95%. As reference, a value of 1 for standard deviation of random effects corresponds to substantial heterogeneity; therefore, a value of 2 is a conservative choice that should accommodate ample between-study variability. Heterogeneity across studies was assessed using the I^2 statistic [9]. We conducted sensitivity analyses to investigate the robustness of our results to various assumptions to our choice of prior distribution for the between-study variability. In particular, a half-normal distribution with the same property of our initial prior was used. Furthermore, a sensitivity analysis was performed for outcomes with very few events (disabling stroke, all-cause mortality, and coronary obstruction).

3. Results

Included were 3 Food and Drug Administration (FDA)-approved investigational device exemption (IDE) trials that enrolled a total of 1810 low-risk patients for TAVR. These three trials were LRT (Low Risk TAVR), PARTNER (Placement of AoRTic TraNscathetER Valve) 3, and Evolut Low Risk. The included trials compared TAVR to SAVR for the treatment of symptomatic severe tricuspid AS in low-risk patients [1–3]. Additionally, these trials each included enrollment into a separate arm of patients with bicuspid AS who underwent TAVR [4,10,11]. The design and baseline clinical characteristics of all three trials are outlined in Table 1. In brief, a total of 1430 low-risk patients with tricuspid AS and

Table 1
Baseline characteristics of included trials.

Clinical trial	Aortic valve morphology	Size (n)	Baseline demographics			Bicuspid morphology		
			Age (years)	Male (%)	STS (%)	Sievers 0 (%)	Sievers 1 (%)	Sievers 2 (%)
PARTNER 3	Bicuspid	71	68.5	69.0	1.4	14.1 (10/71)	84.5 (60/71)	1.4 (1/71)
	Bicuspid CAP	98	71.7	44.9	1.5			
	Tricuspid	496	73.3	67.5	1.9	–	–	–
Evolut low-risk	Bicuspid	150	70.3	52.0	1.4	9.3 (14/150)	90.7 (136/150)	0.0 (0/150)
	Tricuspid	734	74.0	63.8	1.9	–	–	–
LRT	Bicuspid	61	68.6	42.6	1.5	13.1 (8/61)	77.0 (47/61)	9.9 (6/61)
	Tricuspid	200	73.6	61.5	1.8	–	–	–
Total	Bicuspid	380				11.3 (32/282)	86.2 (243/282)	2.5 (7/282)
	Tricuspid	1430				–	–	–

STS, Society of Thoracic Surgeons; PARTNER, Placement of AoRtic TraNscathetER Valves; CAP, continued access protocol; LRT, Low Risk TAVR.

380 low-risk patients with bicuspid AS underwent TAVR. Among those with bicuspid AS and available Sievers classification, 11.3% ($n = 32$) were Sievers 0, 86.2% ($n = 243$) were Sievers 1, and 2.5% ($n = 7$) were Sievers 2 (Table 1). Specific bicuspid anatomy was not available from the continued-access-protocol arm of the PARTNER 3 trial. The BIVOLUT-X registry was not included in our analysis, as it was not performed in the US, did not include a prospective bicuspid arm for comparison, and included a mixture of low- and intermediate-risk patients.

In the Evolut Low Risk trial, patients underwent TAVR using either the second-generation CoreValve Evolut R (Medtronic, Minneapolis, Minnesota) or the third-generation CoreValve Evolut PRO (Medtronic) [2]. Patients in the PARTNER 3 trial received the third-generation Edwards SAPIEN 3 (Edwards Lifesciences, Irvine, California) valve [1]. For tricuspid patients in the LRT trial, self-expanding valves were used in 11.8% of patients, and balloon-expandable valves were used in 88.2% of patients [3]. For bicuspid patients in the LRT trial, self-expanding valves were used in 26% of patients and balloon-expandable valves were used in 74% of patients [4]. Among all patients with tricuspid AS, 52.9% ($n = 757$) received a self-expanding valve and 47.1% ($n = 673$) received a balloon-expandable valve. In those with bicuspid AS, 43.7% ($n = 166$) received a self-expanding valve and 56.3% ($n = 214$) received a balloon-expandable valve.

The overall rates of adverse events were very low in both arms (Fig. 2). There were no significant differences in 30-day all-cause mortality (OR, 0.38; 95% CrI: 0.08, 1.78; I^2, 0%), non-disabling stroke (OR, 0.45; 95% CrI: 0.15, 1.07; I^2, 9%), major vascular complications (OR, 1.45; 95% CrI: 0.63, 3.74; I^2, 0%), or permanent pacemaker implantation (OR, 0.86; 95% CrI: 0.41, 1.47; I^2, 59%). There were statistically significant differences in disabling stroke (OR, 0.16; 95% CrI: 0.01, 0.90; I^2, NA) and coronary obstruction (OR, 0.21; 95% CrI: 0.05, 0.91; I^2, 0%) that disappeared after sensitivity analysis by adding a single event to the tricuspid arm. The sensitivity analyses can be found in the Supplementary Appendix, Supplementary Fig. S1.

With regard to hemodynamic outcomes for patients with reported echocardiographic data at 30 days, there were no differences in MG (OR, 0.00; 95% CrI: −1.46, 1.13; I^2, 81%), AVA (OR, −0.07; 95% CrI: −0.21, 0.07; I^2, 0%), or >mild PVL (OR, 1.08; 95% CrI: 0.11, 5.90; I^2, 45%). (Fig. 3).

4. Discussion

In the present systematic review and meta-analysis of data from 3 prospective trials of low-risk bicuspid and tricuspid patients undergoing TAVR, overall rates of disabling stroke and coronary obstruction were extremely low, with no significant differences in all-cause mortality, non-disabling stroke, major vascular complications, or permanent pacemaker implantation at 30 days. Furthermore, hemodynamics in bicuspid aortic valve patients were comparable with tricuspid aortic valve patients at 30 days. This meta-analysis is the first to compare short-term outcomes in low-risk patients with bicuspid versus tricuspid AS undergoing TAVR incorporating data from the FDA-approved IDE trials in the US.

Randomized trials comparing TAVR to SAVR have excluded bicuspid AS patients [1,2]. There have been concerns regarding their non-circular annuli leading to more PVL and an increased risk for annular rupture. Furthermore, there is concern that severe calcification in these patients may contribute to higher rates of peri-procedural stroke, according to data from a prospective registry on low- and intermediate-risk patients who underwent TAVR using self-expanding valves [12]. However, data on bicuspid patients from the Society of Thoracic Surgeons (STS)/American College of Cardiology (ACC) Transcatheter Valve Therapy (TVT) Registry have previously demonstrated mixed results. A retrospective propensity-score-matched analysis from the STS/ACC TVT registry of 2726 bicuspid patients who underwent TAVR using the SAPIEN 3 valve between 2015 and 2018 reported no differences in 30-day or 1-year mortality, but patients with bicuspid AS had an increased 30-day risk of stroke [13]. A second retrospective propensity-score-matched analysis from the STS/ACC TVT registry of 932 bicuspid patients who underwent TAVR using the Evolut R or Evolut PRO valve between 2015 and 2018 demonstrated no differences in 30-day or 1-year mortality, stroke, or valve hemodynamics [14]. This led to the FDA approving a revised commercial labeling for TAVR last year, modifying a previous precaution for the treatment of bicuspid AS in low-risk patients [15].

In each of the prospective low-risk trials included in this analysis, there were separate prospective bicuspid registries designed to further investigate the concerns raised by observational data in the STS/ACC TVT registry. In our systematic review, clinical outcomes at 30 days comparing low-risk bicuspid patients to low-risk tricuspid patients undergoing TAVR were reassuring. There were no significant differences in all-cause mortality, non-disabling stroke, major vascular complications, permanent pacemaker implantation, or coronary obstruction at 30 days. Our sensitivity analysis of disabling stroke and coronary obstruction demonstrated that with such a low number of events, the results can be highly variable and should be interpreted with caution. With regard to hemodynamics, there were no significant differences between the two groups at 30 days, with no differences in MG, AVA, or >mild PVL.

One of the main observations in this study is that disabling stroke was incredibly rare in low-risk patients, both tricuspid and bicuspid, and this is an encouraging finding. We did observe a statistically significant increase in disabling stroke in bicuspid AS; however, it is important to point out that the results were based on only 4 events. With so few events, the results must be interpreted with caution. When performing a sensitivity analysis, we observed that the addition of even just one event in the tricuspid arm results in an insignificant difference between tricuspid and bicuspid patients. Thus, we believe that the difference observed in the data should be examined in a larger study. In addition, in most patients in these trials, cerebral embolic protection devices were not used. These devices, if routinely used in the bicuspid patients, may mitigate the difference of disabling stroke. Ultimately, more prospective data are needed before TAVR can be widely accepted as the gold standard for the treatment of bicuspid AS.

In the trials examined, there was a relatively even distribution of bicuspid patients receiving a self-expanding valve ($n = 166$) and patients receiving a balloon-expandable valve ($n = 214$). However, the bicuspid anatomic distribution was much less uniform, with Sievers 1 and 2 anatomy comprising 88.7% ($n = 250$) of the bicuspid cohort. Only 32 patients included in this analysis had Sievers 0 bicuspid anatomy; therefore, the results of this analysis cannot necessarily be extrapolated to patients with true bicuspid AS. It is important to note that in the STS/ACC TVT registry, bicuspid anatomy is not specified. Therefore, we cannot be sure of the precise distribution of Sievers bicuspid anatomy in those retrospective analyses. In a true bicuspid patient, Sievers 0, the leaflets lack raphe, resulting in a more elliptical annulus, presenting numerous challenges for valve sizing and deployment. This has raised concerns for annular rupture and PVL. Thus, we must consider Sievers 0 a separate entity from Sievers 1/2.

This study has limitations. Sample size in the bicuspid arm of each trial examined limits our analysis and interpretation of the results. Certain outcomes had so few events (all-cause mortality, disabling stroke, or coronary obstruction at 30 days), that the results are not clinically relevant. More data are needed to elucidate the true differences, if any, between tricuspid and bicuspid patients. We did not have access to individual patient data from the prospective trials analyzed. As such, a multivariate or subgroup analysis was not possible, and we could not perform a propensity-matched analysis between bicuspid and tricuspid cohorts. Nonetheless, individual patient data could help identify the subset of bicuspid patients who could benefit most from TAVR. Additionally, the results are only applicable to low-risk patients eligible for participation in these trials. Finally, the overwhelming majority of patients had Sievers 1 or 2 bicuspid anatomy; thus, the results of these trials do not apply to patients with Sievers 0 bicuspid anatomy, which are a lower-risk and younger population with different anatomical considerations.

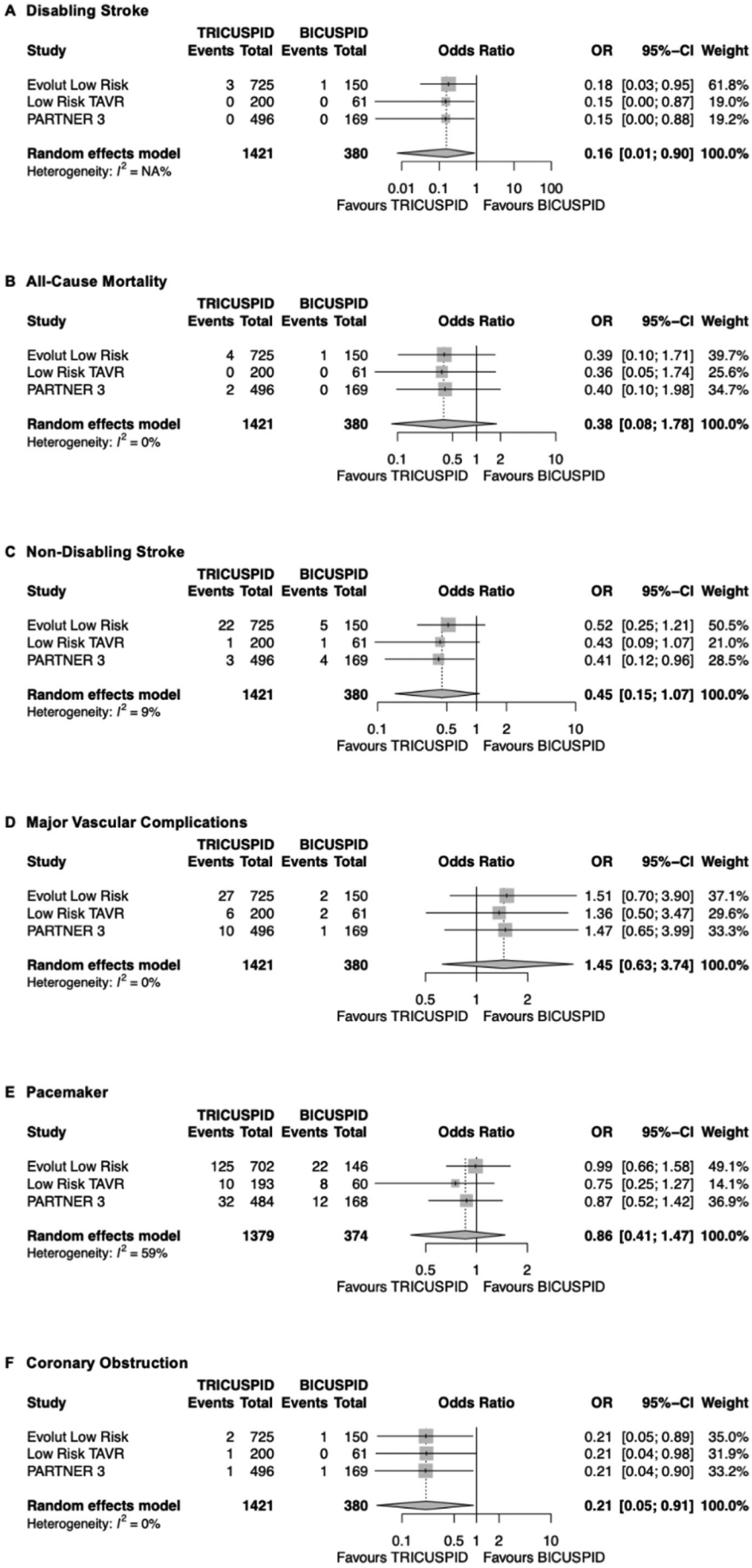

Fig. 2. 30-day clinical outcomes.

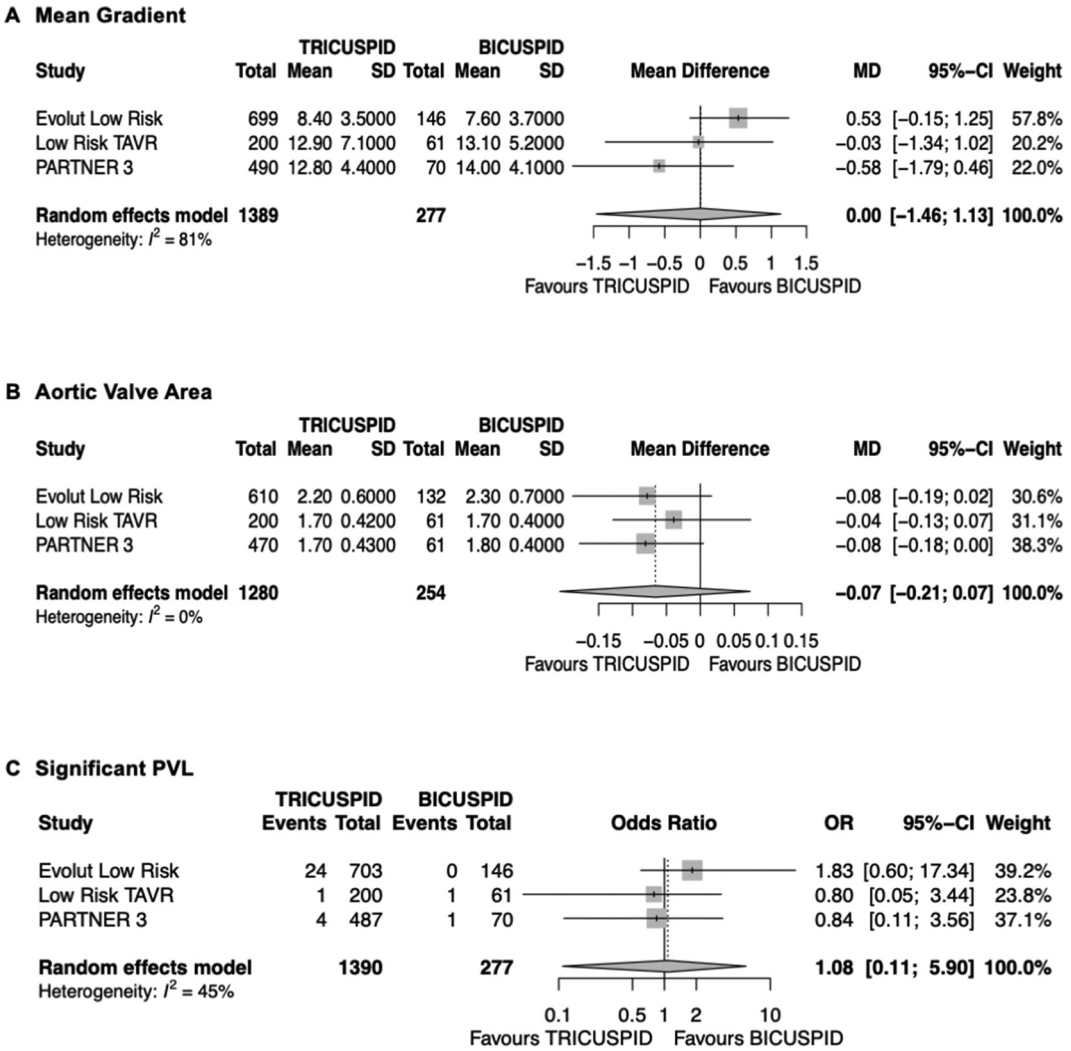

Fig. 3. 30-day hemodynamic outcomes.

5. Conclusion

TAVR in low-risk bicuspid patients resulted in low rates of all-cause mortality, disabling stroke and coronary obstruction, with similar short-term hemodynamics, comparable to low-risk tricuspid patients. Sievers type 0 bicuspid valve morphology represented only a minority of patients in the included studies. More data are required to determine whether patients with severe bicuspid AS undergoing TAVR are at higher risk for stroke.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.carrev.2021.06.123.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Giorgio A. Medranda: Conceptualization, Data curation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Toby Rogers:** Conceptualization, Supervision, Writing – review & editing. **Gheorghe Doros:** Formal analysis, Methodology, Writing – review & editing. **Cheng Zhang:** Formal analysis, Methodology, Writing – review & editing. **Brian C. Case:** Data curation, Methodology, Validation, Writing – review & editing. **Charan Yerasi:** Writing – review & editing. **Christian C. Shults:** Writing – review & editing. **Ron Waksman:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Toby Rogers – Proctor and Consultant: Medtronic, Edwards Lifesciences; Advisory Board: Medtronic; Equity interest: Transmural Systems.

Ron Waksman – Advisory Board: Abbott Vascular, Amgen, Boston Scientific, Cardioset, Cardiovascular Systems Inc., Medtronic, Philips, Pi-Cardia Ltd.; Consultant: Abbott Vascular, Amgen, Biotronik, Boston Scientific, Cardioset, Cardiovascular Systems Inc., Medtronic, Philips, Pi-Cardia Ltd., Transmural Systems; Grant Support: AstraZeneca, Biotronik, Boston Scientific, Chiesi; Speakers Bureau: AstraZeneca, Chiesi; Investor: MedAlliance; Transmural Systems.

All other authors – None.

Acknowledgments

The authors acknowledge Jason Wermers for editing assistance.

References

[1] Mack MJ, Leon MB, Thourani VH, Makkar R, Kodali SK, Russo M, et al. Transcatheter aortic-valve replacement with a balloon-expandable valve in low-risk patients. N Engl J Med. 2019;380:1695–705.

[2] Popma JJ, Deeb GM, Yakubov SJ, Mumtaz M, Gada H, O'Hair D, et al. Transcatheter aortic-valve replacement with a self-expanding valve in low-risk patients. N Engl J Med. 2019;380:1706–15.

[3] Waksman R, Rogers T, Torguson R, Gordon P, Ehsan A, Wilson SR, et al. Transcatheter aortic valve replacement in low-risk patients with symptomatic severe aortic stenosis. J Am Coll Cardiol. 2018;72:2095–105.

[4] Waksman R, Craig PE, Torguson R, Asch FM, Weissman G, Ruiz D, et al. Transcatheter aortic valve replacement in low-risk patients with symptomatic severe bicuspid aortic valve stenosis. JACC Cardiovasc Interv. 2020;13:1019–27.

[5] Ramlawi B. Primary results from the Evolut low risk bicuspid study. Available from: https://www.acc.org/clinical-topics/~/media/2A8C008871314495A4686B0F428D5374.pdf. [Accessed 30 October 2020].

[6] Williams M. The PARTNER 3 trial bicuspid registry for SAPIEN 3 TAVR in low-risk patients. Available from: https://www.tctconnect.com/cme-program-transcatheter-aortic-valve-replacements-for-bicuspid-aortic-valves-sponsored-by-crf-and-supported-through-an-unrestricted-educational-grant-by-edwards-lifesciences/. [Accessed 30 October 2020].

[7] Hutton B, Salanti G, Caldwell DM, Chaimani A, Schmid CH, Cameron C, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. Ann Intern Med. 2015;162:777–84.

[8] Plummer M. JAGS: a program for analysis of Bayesian graphical models using Gibbs sampling. Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003). Vienna, Austria; 2003. p. 1–10.

[9] Spiegelhalter DJ, Abrams KR, Myles JP. Bayesian approaches to clinical trials and health-care evaluation. John Wiley & Sons, Ltd; 2004.

[10] Ramlawi B. Primary results from the Evolut low risk bicuspid study. https://www.acc.org/clinical-topics/~/media/2A8C008871314495A4686B0F428D5374.pdf. [Accessed 30 October 2020].

[11] Williams M. The PARTNER 3 trial bicuspid registry for SAPIEN 3 TAVR in low-risk patients. https://www.tctconnect.com/cme-program-transcatheter-aortic-valve-replacements-for-bicuspid-aortic-valves-sponsored-by-crf-and-supported-through-an-unrestricted-educational-grant-by-edwards-lifesciences/. [Accessed 30 October 2020].

[12] Tchetche D. Transcatheter treatment of bicuspid aortic valves with the Evolut platform: the BIVOLUT-X registry. https://media.pcronline.com/diapos/PCReCourse2020/254-20200626_1504_Hotline_and_Innovation_Channel_Tchetche_Didier_0000_(557)/Tchetche_Didier_20200626_1440_Hotline_and_Innovation_-_Channel_3.pdf. [Accessed 30 October 2020].

[13] Makkar RR, Yoon SH, Leon MB, Chakravarty T, Rinaldi M, Shah PB, et al. Association between transcatheter aortic valve replacement for bicuspid vs tricuspid aortic stenosis and mortality or stroke. JAMA. 2019;321:2193–202.

[14] Forrest JK, Kaple RK, Ramlawi B, Gleason TG, Meduri CU, Yakubov SJ, et al. Transcatheter aortic valve replacement in bicuspid versus tricuspid aortic valves from the STS/ACC TVT registry. JACC Cardiovasc Interv. 2020;13:1749–59.

[15] Medtronic Evolut TAVR system receives updated U.S. labeling revising precaution for treatment of low-risk patients with bicuspid aortic stenosis. https://newsroom.medtronic.com/news-releases/news-release-details/medtronic-evolut-tavr-system-receives-updated-us-labeling. [Accessed 27 January 2021].